Bayesian Network Modeling of

NFL Game Stats & Market Signals

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3,990 NFL Games	15 Seasons
6 Market Pts / Game	2 Data Sources

The Question, Before the Math

Does any of this tell you something the box score doesn't?

NFL · CLOSING LINES — illustrative

MATCHUP	SPREAD (OPEN → CLOSE)		TOTAL	MONEYLINE
IND vs NO	-5.0 → -5.5		52.5	-240
PIT vs CLE	-4.5 → -5.0		36.5	-235
GB vs PHI	-2.5 → **-3.0**	LINE MOVE	43.0	-150
DEN vs BUF	-3.0 → -2.5		38.5	-130

Opening line → closing line. The **movement** is the signal we test against the box score.

Objectives

Is line movement **conditionally independent**
of ATS outcome given game fundamentals?

Objectives

- Test whether line movement is conditionally independent of ATS outcome given game fundamentals
- Isolate whether market signals add predictive information beyond game statistics

Our models proved that market signals:

- **do NOT** improve accuracy beyond game stats
- **but they DO** improve calibration

Gap in the Literature

BNs for Sports Prediction

Constantinou, Fenton & Neil (2012–13): BN for soccer forecasting — game variables only, no market movement.

Stats + Odds Combined

Egidi, Pauli & Torelli (2018): hierarchical Bayes blending odds — no conditional dependence structure.

Market Efficiency

Simon (2024), Moskowitz (2021): line-movement efficiency — no game-stats integration.

GAP CONFIRMED

Methodology

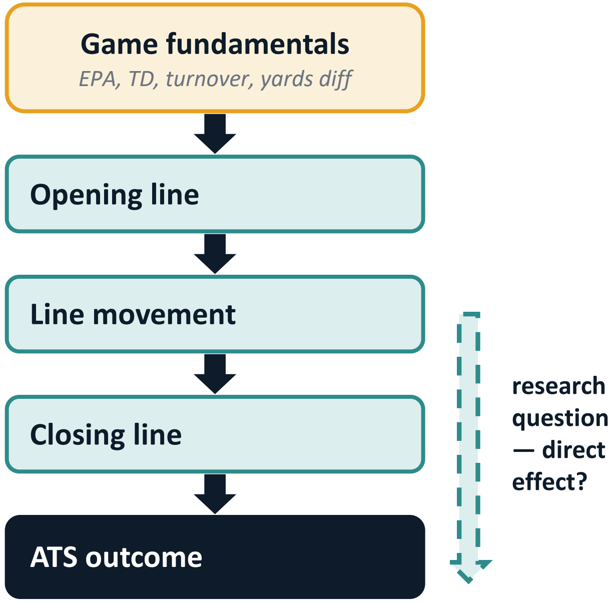
MODEL ARCHITECTURE

- Naive Bayes baseline — all features independent
- Hill Climbing BN — score-based learning
- PC Algorithm BN — constraint-based check

EVALUATION

- Accuracy — hard prediction performance
- Log Loss — probability calibration
- Target: Cover / Loss / Push (3-class)

ASSUMED CAUSAL STRUCTURE



Ablation: 3 feature sets (Stats · Market · Combined) × 3 models (NB · HC · PC) = 9 configurations, each scored on accuracy and log loss.

The Result Is Not What You'd Expect

ACCURACY

Market signals HURT

84.1%

Stats-only Naive Bayes

81.0%

Combined (stats + market)

Adding market features lowers accuracy.

VS

CALIBRATION

Market signals HELP

0.345

Combined PC log loss (best)

0.439

Stats-only NB log loss

Adding market features improves calibration.

Results: Accuracy vs. Calibration

ACCURACY

84.1%

Stats-only Naive Bayes is the highest of any configuration. Adding market features lowers NB accuracy to 81.0%; both Bayesian Networks stay at or below the 58.0% majority baseline in every configuration.

CALIBRATION (LOG LOSS)

0.345

Combined PC algorithm — best log loss of all nine runs. Market signals sharpen probabilities without changing hard predictions.

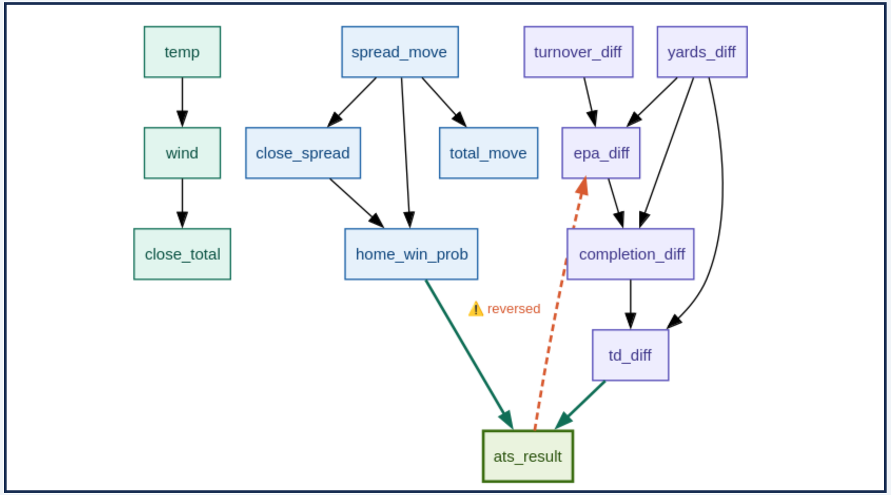
Config	NB Acc	HC Acc	PC Acc	NB Loss	HC Loss	PC Loss
Stats-Only	**84.1%**	50.4%	50.4%	0.439	0.375	0.368
Market-Only	56.4%	58.6%	58.6%	0.840	0.776	0.760
Combined	81.0%	50.4%	47.3%	0.538	0.375	**0.345**

CENTRAL FINDING

Market signals improve how certain the model is — not what it predicts.

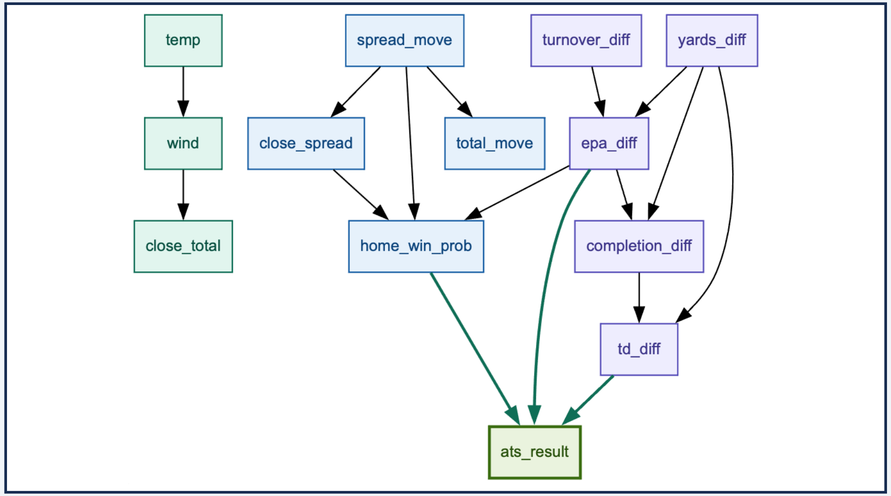
The Model Made a Causal Error — a Second Algorithm Caught It

HILL CLIMBING — raw learned DAG



Hill Climbing learned ats_result → epa_diff — a betting outcome cannot cause the on-field play that produced it. Causally impossible.

PC ALGORITHM — independent result



Run independently, the PC algorithm recovered epa_diff → ats_result on its own — with home_win_prob and close_spread as direct parents. Not confirmation bias: a second method found the same structure.

The algorithm made a causal error — and we caught it:
domain knowledge flipped the edge, and a second, independent algorithm (PC) recovered the same direction on its own.

Implementation

DATA PIPELINE

- nflverse (game stats) + SportsBookReviewsOnline (market data)
- 3,990 games, 2007–2022, joined by season + matchup
- Game Stats: EPA, TD, turnover, yards differentials
- Market Signals: spread & total movement, home win prob, rolling ATS

MODELS

- Gaussian Naive Bayes => continuous features, no discretization
- Hill Climbing BN + PC Algorithm BN => 3-bin ordinal encoding
- 80/20 train-test split

VALIDATION

- Dual-algorithm approach: HC and PC run independently and cross-checked
- HC reversed edge corrected via domain knowledge, confirmed by PC

Challenges & Limitations

Retrospective Statistics

EPA and other game stats are post-game measurements. Stats-only accuracy reflects information content, not pre-game predictive power.

Discretization Penalty

BNs require binning continuous features; magnitude information is lost. The main reason BNs trail Naive Bayes on accuracy.

Class Imbalance

Push is only 3.0% of the data. Neither BN ever predicts it; NB reaches just 22% F1 on the class.

Spread Direction

We bin the closing spread by magnitude, which discards direction: a home team laying seven and one receiving seven fall in the same bin. Recovering signed direction is left to future work.

Future Work & Conclusions

FUTURE WORK

- Continuous (Gaussian) BN formulations to eliminate the discretization penalty
- Post-2018 PASPA analysis only — isolate the modern legal betting market
- Recover spread direction; revalidate the target variable
- Add quarterback injury data as a high-signal feature

CONCLUSIONS

- Partial efficiency confirmed: the market's hard prediction is already encoded in game stats
- Calibration finding: the market's uncertainty estimate adds information beyond statistics alone
- PC + HC dual-algorithm design caught and corrected a causally impossible edge
- A nuanced answer that advances both sides of the efficiency debate

QUESTIONS & DISCUSSION

For accuracy: the market adds no signal beyond game stats.

For calibration: the market's uncertainty estimate does.

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